

Yuanchao Xu

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Room 4-405, Maskawa Building for Education and Research, Kyoto University

June 29, 2026

Working Experience

University of Alberta

Visiting Postdoc, Applied Mathematics
April 2026 - August 2026

Kyoto University

Postdoc, SACRA
September 2025 - August 2030

Education Experience

University of Alberta

Doctor of Philosophy (*Dissertation Award*), Applied Mathematics
September 2021 - August 2025

University of Manitoba

Master of Science, Mathematics
September 2018 - August 2020

University of Oregon

Bachelor of Science (*Cum Laude*), Mathematics
January 2015 - December 2017

Research Interest

My research interest lies in data-driven methods for stochastic dynamical system. In particular, I am interested in building up theoretical framework for learning stochastic Koopman operator and applying it into complex real-world problems, e.g., computational neuroscience, reinforcement learning and generative modeling, etc.

Research

1. Neural Operators Meet Kolmogorov Backward Operators: A Variational Framework for Spectral Learning and Forecasting (In preperation, 2025) (*1st author*)
2. Generative Modeling through Koopman Spectral Analysis: An Operator-Theoretic Perspective (arXiv, 2025) (*1st author*)
3. Reinforcement-learning-guided data-driven estimation of spectral properties of stochastic Koopman semigroups (DCDS-I, 2025) (*1st author*)
4. Spectral analysis of Koopman operator through pseudo-resolvent ((Submitted to Physica D) arXiv, 2025) (*1st author*)
5. A Data-Driven Framework for Koopman Semigroup Estimation in Stochastic Dynamical Systems (Chaos, 2025) (*1st author*)
6. Koopman Eigenfunctions Links Multiscale State-Dependent Brain Dynamics (In preperation, 2025)
7. ResKoopNet: Learning Koopman Representations for Complex Dynamics with Spectral Residuals (ICML2025, Vancouver, Canada, 2025) (*1st author*)
8. Koopman Spectral Analysis Uncovers the Temporal Structure of Spontaneous Neural Events (COSYNE (poster), Lisbon, Portugal, 2024)
9. Decentralized Multi-Agent Reinforcement Learning for Task Offloading Under Uncertainty (arXiv, 2021) (*1st author*)

Talk

1. *Generative Modeling through Koopman Spectral Analysis*
Massachusetts Institute of Technology, Cambridge, MA, April 27, 2026
2. *Generative Modeling through Koopman Spectral Analysis*
CMS Special Seminar
California Institute of Technology, Pasadena, CA, April 13, 2026

3. *Generative Modeling through Koopman Spectral Analysis*
Probability Seminar
University of Oregon, Eugene, OR, April 8, 2026
4. *Generative Modeling through Koopman Spectral Analysis*
MLDS 5 & DEDS 2026
Kyoto University, Kyoto, Japan, February 9-13, 2026
5. *Generative Modeling through Koopman Spectral Analysis*
Machine Learning and Dynamical Systems Seminar
Alan Turing Institute, London, UK, January 8, 2026 (Online)
6. *Data-Driven Koopman Framework for Modeling Complex Dynamical Systems*
Fudan University, Shanghai, China, January 5, 2026
7. *Data-Driven Koopman Framework for Modeling Complex Dynamical Systems*
European Molecular Biology Laboratory, Rome, Italy, November 24, 2025 (Online)
8. *Data-Driven Koopman Framework for Modeling Complex Dynamical Systems*
Laboratoire Hubert Curien, Saint-Étienne, France, October 14, 2025
9. *Perturbation method for learning stochastic Koopman operator*
CREST
RIKEN, Kobe, Japan, December 26-28, 2024
10. *Extracting Dynamics from Complex Systems: Deterministic and Stochastic Perspectives*
East China University of Science and Technology, Shanghai, China, December 19, 2024

Visiting

1. Massachusetts Institute of Technology, Cambridge, MA, April 27-30
2. Cornell University, Ithaca, NY, April 16-18
3. California Institute of Technology, Pasadena, CA, April 13-15
4. University of California, Santa Barbara, CA, April 10th
5. University of Oregon, Eugene, OR, April 6-9
6. Fudan University, Shanghai, China, January 5-9, 2026
7. Laboratoire Hubert Curien, Saint-Étienne, France, October 14-15, 2025
8. European Molecular Biology Laboratory, Rome, Italy, October 11-13, 2025
9. Istituto Italiano di Tecnologia, Genoa, Italy, October 9-10, 2025
10. Ehime University, Matsuyama, Japan, December 23-25, 2024

Reviewer

ICML2026, ICLR2026, IEEE-TNNLS, IEEE-TETCI

Service

Young Associate Editor: DCDS-I

Awards

- Faculty of Science Doctoral Dissertation Award, 2025
- Mary Louise Imrie Graduate Student Award, 2025
- Dr. Josephine M. Mitchell Recruitment Scholarship, 2021
- International Graduate Student Entrance Scholarship, 2020
- International Graduate Student Entrance Scholarship, 2018
- Clarence and Lucille Dunbar Scholarship, 2017

**Teaching
Experience**

<i>Math 102(lab) Applied Linear Algebra</i>	<i>Winter 2025</i>
<i>Math 209(lab) Calculus for Engineering 3</i>	<i>Fall 2024</i>
<i>Math 209(lab) Calculus for Engineering 3</i>	<i>Spring 2024</i>
<i>Math 102(lab) Applied Linear Algebra</i>	<i>Winter 2024</i>
<i>Math 201(lab) Differential Equations</i>	<i>Fall 2023</i>
<i>Math 101(lab) Calculus for Engineering 2</i>	<i>Spring 2023</i>
<i>Math 201(lab) Differential Equations</i>	<i>Winter 2023</i>
<i>Graduate Research Assistant</i>	<i>September 2020 - August 2021</i>
<i>Graduate Teaching Assistant</i>	<i>September 2018 - August 2020</i>
<i>Undergraduate Teaching Assistant</i>	<i>September 2016 - December 2017</i>
<i>Undergraduate Math Tutor</i>	<i>April 2016 - December 2017</i>